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Precision Cutting and Multi-Stage Fishing Operation Using a Tailor-Made E-line Mechanical Cutter in a Restricted ID Mandrel

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Abstract

This paper documents a first-of-its-kind electric-line intervention to segment and recover a straddle assembly lodged above a downhole safety valve within a restricted internal-diameter (ID) mandrel. Conventional jarring and high-force retrieval failed due to scale-induced confinement and persistent slip engagement. A bespoke slim mechanical cutter head (≈ 1.7 -in. OD) was engineered to operate inside a 1.811-in. ID, with a tell-tale/no-go feature enabling a precise cut 975 mm (≈ 37.5 in.) from the inner sleeve top while minimizing torque and swarf generation.

Full-scale surface integration tests calibrated cutting pressure and verified reliability before offshore deployment. The downhole cut was executed in ≈ 5 h with real-time SRO monitoring. The straddle was then recovered in three stages using an electric-line hydraulic stoker and actuated grapple: 77 klbf for the upper section, 70 klbf to pass a weak point, and a final 20/70 klbf push/pull sequence to free the lower body. A downhole video camera provided rapid diagnostics and offshore/onshore decision support. The operation eliminated explosives, reduced risk, and restored full well accessibility. Design-to-deployment was completed in <4 weeks.

The case offers engineering guidance on tool miniaturization, cut positioning, force staging, and debris control for restricted-ID interventions, and demonstrates how agile custom tooling can displace higher-risk methods in time-critical well integrity scenarios.

Introduction

Well A is an oil producer, completed in 1994, has faced recurring integrity challenges throughout its life cycle, culminating in a failed inflow test across the downhole safety-valve (DHSV) envelope. A remedial straddle was installed and initially restored integrity but later failed verification. Multiple recovery attempts using coiled-tubing bottomhole assemblies (BHAs) and jarring were unsuccessful, largely because of a restricted internal diameter (ID) at the tubing hanger and scale-related radial confinement. Earlier coiled-tubing runs suggested the straddle could be raised to the hanger depth but could not pass through an ≈ 0.10 -in. (≈ 2.5 -mm) ID reduction between the tubing hanger and production tubing. Limited surface feedback during these operations reinforced the assumption that the straddle was stationary at the hanger with its fish-neck accessible for engagement.

Despite these efforts, only the fishing neck was recovered, leaving the main body lodged in the well. A comprehensive options review concluded that conventional force-based retrieval was unlikely to succeed and led to the adoption of an internal-mandrel segmentation strategy: perform a precision internal cut at a controlled depth to release slip loading, followed by staged recovery using electric-line mechanical tools. This paper presents the technical basis, tool development, execution steps, and key outcomes of this approach.

Statement of Theory and Definitions

Overview of well intervention theory

Well interventions diagnose, maintain, or enhance well performance across the life cycle. Selecting the conveyance method depends on well architecture (IDs, deviations, restrictions), intervention objective (diagnostics, manipulation, cutting/fishing, circulation/milling), barrier philosophy, and surface logistics. In practice, Slickline (SL), Electric Line (EL), and Coiled Tubing (CT) are often combined to balance precision, force, and fluid capability

Slickline is a single-strand, non-conductive wire used to deploy mechanical tools into the wellbore without real-time telemetry. It is widely applied for setting and retrieving plugs and flow-control devices, shifting sleeves, and performing basic mechanical manipulations. Slickline is also used to run drifts and gauge cutters, tag no-go profiles, and provide basic diagnostic confirmation such as identifying the top of fish or detecting restrictions. Its simplicity, low cost, and minimal surface footprint make it ideal for routine maintenance and diagnostic tasks in wells with known geometry and low complexity. The strengths and limitations of which are summarized in [Table 1](#).

Table 1—Strengths and Limitations of Slickline Method of Well Intervention

Strengths	Limitations
Simple, low-footprint, rapid mobilization	No SRO (Surface Read-Out) telemetry; limited situational awareness
Minimal surface power/equipment; efficient for routine tasks	Limited axial force (typically <~2 klbf without specialized aids)
	Conveyance challenges in high deviation without tractor assistance

Electric line, or e-line, is a multi-conductor cable that enables deployment of powered mechanical tools while providing real-time surface read-out (SRO) for depth, tension, current, and stroke. This capability allows precise depth control and dynamic decision-making during operations. Typical applications include diagnostic logging such as multifinger caliper surveys, leak detection, and downhole video inspections. E-line also supports powered mechanical tools like cutters and hydraulic stokers, as well as actuated fishing or grapple tools that provide positive latch confirmation. These features make e-line the preferred choice for high-precision interventions, especially in restricted-ID environments. The strengths and limitations of which are summarized in [Table 2](#).

Table 2—Strengths and Limitations of Electric Line Method of Well Intervention

Strengths	Limitations
Precision control and real-time diagnostics for decision-making	More complex surface equipment/power; cable-handling risks in high deviation
High, anchored axial force via hydraulic stoker (fielded here up to ~100 klbf in staged ramps)	Limited native fluid-pumping capability (circulation typically via well fluids or auxiliary means)
Ideal for precision cutting and segmented fishing in restricted IDs (e.g., internal mandrel cuts)	

EL-based cutting and fishing in restricted IDs is well documented (Movahed, et al. 2023), showing the value of anchored force application and telemetry-driven confirmation to minimize risk and re-runs.

Coiled tubing is a continuous steel tube that allows fluid circulation while conveying mechanical tools into the well. It is commonly used for milling and dressing operations, heavy clean-outs, and fluid placement tasks such as acidizing and cementing. Additional applications include jetting, scale removal, and mechanical manipulation in highly deviated or horizontal wells. The ability to pump fluids while working makes CT particularly effective for debris management and complex interventions where circulation and mechanical force are both required. The strengths and limitations of which are summarized in Table 3.

Table 3—Strengths and Limitations of Coiled Tubing Method of Well Intervention

Strengths	Limitations
Pumping while working; circulation for debris control and pressure management	Larger footprint and cost; requires PCE (Pressure-Control Equipment), injector, reel
Better reach and weight-on-bit for milling; compatible with a wide tools portfolio	Logistic/scheduling complexity offshore; reach can be limited in long horizontals without tractors

CT is the contingency of choice when bulk material removal or sustained circulation is required and when EL-only methods cannot guarantee progress due to debris or geometry.

Integrated Intervention Strategies. Complex recoveries rarely succeed with a single conveyance. The most reliable programs combine SL assurance, EL precision, and CT contingency, sequenced by risk:

- Assure access (SL/EL)
 - Run drifts/gauge cutters and no-go prongs to verify IDs, reach, and depth control.
 - Establish mechanical baselines (running/pulling weights) and confirm cut window clearance.
- Diagnose (EL)
 - Use downhole video and MFC to refine the plan (identify obstructions, scale bonding, slip engagement), and run leak detection if required.
- Execute with precision (EL)
 - Anchor above target; perform internal segmentation cuts with slim mechanical cutters designed to minimize torque and swarf.
 - Apply stroked force ramps (e.g., 5-kJbf increments) with actuated grapples, guided by SRO inflections (tension/current/travel) to confirm events (e.g., "cut complete," "fish movement").
- Circulate/remediate if needed (CT)
 - Deploy CT only when debris, scale, or unstable components demand milling or sustained circulation. Keep disconnect contingencies and debris traps ready.

The Challenge of Scale and Wellbore Access

Wellbore access in mature offshore assets is increasingly constrained by aging infrastructure, mechanical limitations, and operational restrictions. Many wells in the North Sea have been in service for more than 25 years and have undergone extensive waterflooding to maintain reservoir pressure and recovery. Prolonged seawater injection—particularly during early campaigns with inconsistent chemical inhibition—has led to elevated water cut, accelerating the onset of scale deposition and corrosion.

Scaling and Corrosion Mechanisms. Scale, primarily calcium carbonate and barium sulfate, accumulates within production tubing, reducing the effective internal diameter (ID) and impeding tool conveyance.

Concurrently, corrosion—both internal (from reservoir and injected fluids) and external (from annular water ingress)—compromises tubing integrity (Thomas og Norman 2023). These degradation mechanisms (illustrated in Fig. 1) result in:

- Equipment degradation: Documented tubing failures due to wall thinning and flow restrictions.
- Well barrier failures: An increasing number of wells have been identified with compromised barrier integrity, requiring remediation to restore full compliance with the operator's Well Integrity Management System (WIMS).
- Reduced reliability and higher intervention frequency: Necessitating scale milling, chemical squeeze treatments (Ali, et al. 2024), and tubing replacements, which increase resource demand and platform logistics complexity.



(a) Tubing Scale in Well-X

(b) Safety Valve Scale in Well-Y

(c) Corrosion in Well-Z

Figure 1—Images of well bore integrity issues in typical North Sea wells.

Mitigation Strategies. Operators typically adopt a multi-pronged approach:

- Chemical inhibition: Continuous or batch scale inhibitor injection via gas lift or downhole delivery.
- Material upgrades: Use of corrosion-resistant alloys (CRA) or GRE-lined tubing during workovers to extend service life in high-water-cut environments.
- Monitoring and diagnostics: Routine wireline caliper surveys and tubing integrity logs to track wall loss and scale deposition for proactive planning.

Access Constraints and Mechanical Challenges. Nominal tubing ID is typically ~4.19 in., but scale, corrosion, and legacy completion hardware can reduce this to as little as 1.5 in., severely limiting standard tool deployment and often requiring slimline or bespoke equipment. Installation of wireline-retrievable subsurface safety valves (WRSSSVs) or straddle assemblies frequently demands pre-milling or scale removal. Formations such as the Upper and Lower Lark are prone to cavings and pack-off events during operations like back-reaming or casing runs, increasing the risk of stuck tools and delays. These risks are mitigated through drift verification runs and specialized milling tools before deploying critical intervention equipment.

Operational Constraints. Offshore platform limitations further complicate interventions. Restricted crane capacity affects equipment mobilization, while simultaneous operations (SIMOPS) introduce scheduling conflicts. Weather-dependent vessel operations and grating conditions on unmanned installations often require pre-intervention modifications and contingency planning.

Global Context. These challenges are not unique to the North Sea. Similar issues have been reported in other mature offshore assets worldwide. For example in fields located in the Atlantic Ocean off the coast of South America (Jordan, et al. 2006).

Down-hole Safety/Surface Controlled Subsurface Safety Valve (SCSSV) Repair Methods

Surface-Controlled Subsurface Safety Valves (SCSSVs) are critical components of well integrity architecture, providing a fail-safe barrier against uncontrolled flow. In mature assets, SCSSV failures are increasingly common due to aging completions, scale buildup, and control-line degradation. Repair strategies must therefore be technically robust and adaptable to the constraints of legacy well architecture.

Conventional SCSSV Repair Approach. The standard remediation pathway for a mechanically failed SCSSV is the installation of a Wireline-Retrievable Subsurface Safety Valve (WRSSSV). This method typically requires:

- Verification of lock-profile compatibility
- Drift confirmation through the tubing
- Deployment of a WRSSSV with an appropriate seal stack and control-line interface

However, in many mature wells, WRSSSV installation is hindered by scale-induced internal diameter (ID) restrictions or mechanical damage to the lock profile. In such cases, alternative solutions are required.

Straddle-Based Remediation. When WRSSSV installation is not feasible, straddle systems offer a viable alternative for restoring well integrity. These systems isolate the failed SCSSV and re-establish a pressure barrier across the compromised zone. The process generally involves:

- Pre-Installation Diagnostics: Inflow testing and caliper logging confirm leak location and severity. Downhole video or multifinger caliper (MFC) logging may be used to identify tubing damage or scale obstructions.
- Tool Preparation and Function Testing: The selected modular straddle system is function-tested on surface. Timer settings, prong engagement, and pressure integrity are verified before deployment.
- Deployment: The straddle is run on slickline or electric line, depending on configuration. A typical toolstring includes a running tool, jars, and a mechanical setting device. The straddle is landed in the no-go profile above the SCSSV and activated via hydraulic or mechanical triggers.
- Post-Installation Verification: Once set, the straddle is inflow-tested to confirm isolation. Barrier verification is performed using pressure bleed-off and annular monitoring. Serial numbers and depth flags are recorded for traceability.

Advanced Applications and Case Histories. In wells with severe ID restrictions or mechanical damage, bespoke straddle systems have been deployed. For example, in the Well A, a custom-engineered slim cutter was used to segment and retrieve a stuck straddle, followed by installation of a new isolation system. Similarly, in other mature wells, failed straddles have been retrieved and replaced despite unknown scale conditions.

Operators have also implemented extended-length straddle systems (up to 3,000 ft) in wells with multiple leak points. These modular systems allow on-site customization based on leak diagnostics and have demonstrated high reliability, with minimal mis-runs reported.

Operational Considerations. Straddle-based repairs require meticulous planning and coordination:

- Barrier Management: Temporary barriers must be maintained during WRSSSV retrieval and straddle installation.

- Tool Compatibility: Toolstring design must account for ID restrictions, scale, and mechanical damage.
- Contingency Planning: Alternate straddle sizes and WRSSSV configurations should be available in case of deployment failure.

Challenges of Straddle Recovery in a Heavily Scaled Well

In Well A, a straddle assembly was installed as a remedial barrier following the failure of a tubing-retrievable surface safety valve (TRSSSV) and unsuccessful attempts to deploy a wireline-retrievable subsurface safety valve (WRSSSV). While the straddle initially restored integrity, it later failed an inflow test, requiring retrieval. Multiple fishing attempts using coiled tubing and various bottomhole assembly (BHA) configurations were unsuccessful. Although the straddle could be moved axially, it could not be extracted because the slips remained engaged during pick-up, creating a mechanical lock that conventional overpull and jarring could not overcome.

Why Conventional Methods Failed. The straddle was designed to anchor securely within the tubing using slips that expand radially to grip the wall. Under normal conditions, these slips disengage when a specific axial force or mechanical trigger is applied. In this case, several factors prevented release:

- Slip embedment: Over time, the slips had bitten into the tubing wall, increasing resistance.
- Scale bridging: Deposits around the slip teeth restricted retraction.
- Tubing deformation: High overpull loads during earlier attempts likely distorted the contact area.
- Thermal and pressure cycling: These conditions may have further seated the slips.

Even after the fishing neck was recovered and the tubing-hanger ID was enlarged by milling, the main body of the straddle remained immobile. The embedded slips acted as a wedge, resisting both upward and rotational forces.

Strategic Shift. Recognizing that force-based recovery was ineffective, the intervention team adopted a precision segmentation strategy. A custom-engineered slim mechanical cutter was deployed to sever the inner mandrel below the slip assembly. This allowed the straddle to be removed in discrete sections, eliminating the need to disengage the slips entirely. The approach required a cutter capable of operating within a restricted ID and generating minimal swarf to avoid debris-related complications.

Key Lessons. This case demonstrates that retrieval failures in mature wells often result from a combination of chemical and mechanical factors. While scale reduced clearance, the dominant barrier was slip engagement under confined conditions. Successful resolution required:

- Accurate diagnostics to confirm the root cause.
- Purpose-built tooling for restricted-ID cutting.
- A staged recovery plan that prioritized control and minimized risk.

Adaptive strategies like this—combining diagnostics, custom tool design, and segmented recovery—are essential for complex interventions in legacy completions.

Electric-Line (E-Line) Fishing Tools and Grapple

Electric-line (e-line) fishing tools are a cornerstone of modern well-intervention strategies, particularly in complex retrieval scenarios where slickline or coiled tubing methods are ineffective or impractical (Movahed, et al. 2023). These tools combine mechanical functionality with real-time surface read-out (SRO), enabling precise control, diagnostics, and force application during downhole operations.

Tool Architecture and Functionality. E-line fishing toolstrings typically consist of the following components:

- An e-line fishing toolstring typically includes the following components:
- Anchor/Stroker Modules: Hydraulically actuated devices that anchor the toolstring within the tubing and apply controlled axial force—up to approximately 100,000 lbf—for pushing or pulling the fish.
- Grapple Modules: Mechanically actuated latching tools designed to engage specific profiles such as fishing necks, mandrel shoulders, or internal diameters. Interchangeable grapple heads allow adaptation to varying fish geometries.
- Cutting Tools: When segmentation is required, mechanical cutters or non-explosive thermal/plasma cutters are deployed to sever the mandrel or tubing.
- Diagnostic Modules: Sensors that provide real-time feedback on tension, torque, stroke, and current, enabling dynamic adjustments during operations.

Grapple Methodology. The grapple system is central to e-line fishing. It is designed to achieve positive engagement with the fish and withstand high axial loads without slippage. The typical sequence includes:

- Pre-run verification: Drift and prong checks confirm tool access and simulate grapple reach.
- Anchoring: The stroker anchors above the target depth to provide a stable base for force application.
- Engagement: The grapple is extended to latch onto the fish profile, confirmed by SRO indicators.
- Force application: Controlled force ramps are applied in increments, monitored via real-time tension and stroke data to detect movement or slippage.
- Contingency: If engagement fails, alternate grapple sizes or cutting tools are deployed.

Applications and Case Studies. E-line fishing tools are particularly effective in:

- Retrieving stuck straddles or WRSSSVs: In Well A, an anchored stroker and grapple system successfully recovered segmented straddle sections after precision cutting of the inner mandrel.
- Fishing slickline tools and plugs: (Dulic 2018) documents the use of an e-line stroker to recover a stuck slickline tool after multiple jarring attempts failed, applying up to 60,000 lbf in a controlled manner.
- Milling and debris removal: Modern e-line milling tools integrate real-time diagnostics and adaptive control of weight-on-bit and penetration rate, improving efficiency in restricted-ID environments (Altarelli, et al. 2024).

Operational Considerations. Successful deployment requires:

- Surface testing and calibration: Verification of anchor stroke, grapple engagement, and cutter extension before deployment.
- Real-time monitoring: Continuous observation of tension, current, and stroke to confirm engagement and cutting progress.
- Contingency readiness: Availability of alternate grapples, cutters, and fishing tools to adapt to unexpected conditions. In Well A, both mechanical and thermal cutters were staged, along with multiple grapple configurations.

Compared to slickline or coiled tubing, e-line fishing tools offer:

- Higher force application: Up to 100,000 lbs of axial force can be applied with precision.

- Real-time diagnostics: Enables immediate feedback and decision-making.
- Compact deployment: Suitable for restricted deck space and limited crane capacity.
- Reduced risk: Controlled force application minimizes the risk of damaging the fish or wellbore.

Cutting Tools in Well Intervention and their Application in Well A

Cutting tools are critical in well-intervention operations where mechanical segmentation of downhole assemblies is required to enable retrieval, isolation, or abandonment. These tools must operate reliably in constrained environments, often under high-pressure, high-temperature (HPHT) conditions—while delivering precision and minimizing debris generation.

Tool Design and Functionality. Modern cutting tools for electric-line (e-line) interventions are generally mechanical or thermal:

- Mechanical cutters use a rotating cutting head to incrementally remove material from the target component. They are engineered to:
 - Operate within restricted internal diameters (e.g., 1.811 in. ID in Well A)
 - Apply minimal torque to reduce stalling or component rotation
 - Generate fine swarf for easier debris management
 - Incorporate telltale and no-go features for accurate depth control
- Thermal or plasma-based cutters employ high-energy plasma to sever tubulars or mandrels. These are typically deployed when mechanical cutters are ineffective due to material hardness, eccentricity, or access limitations.

Deployment and Execution. In Well A, a custom-engineered slim mechanical cutter was deployed to sever the inner mandrel of a stuck straddle assembly. The cutter was anchored above the target zone using a hydraulic stroker and extended to a depth of approximately 37.5 in. from the top of the inner sleeve. Real-time surface read-out (SRO) monitored current, torque, and piston travel throughout the operation.

Key execution steps included:

- Pre-run drift checks to confirm access and cutter reach
- Surface validation of cutting pressure and tool response
- Continuous circulation of inhibited seawater to manage swarf and pressure spikes
- Post-run inspection of cutter arms and telltale travel to confirm cut completion

The cut enabled the straddle to be segmented and retrieved in multiple stages, restoring full well access without explosives. The strengths and limitations are illustrated in [Table 4](#).

Table 4—Strengths and Limitations of Deploying an Electric Line Cutting Tool

Strengths	Limitations
Precise, controlled cuts with minimal collateral damage	Limited effectiveness in highly eccentric or rotating components
Real-time monitoring and diagnostics	Potential for tool stalling in high-debris environments
Compatibility with restricted-ID environments	Requires accurate depth control and pre-job planning
Reduced operational risk compared to explosive methods	

Application in Well A. Following the successful internal cut, the segmented straddle was recovered in two phases:

- Upper Section Retrieval: The grapple tool latched onto the top sub and retrieved the upper segment with approximately 33,000 lbf of force.
- Lower Section Retrieval: After a magnet run to remove residual slips and dogs, the same stroker-grapple configuration was redeployed to engage and recover the lower mandrel body, with careful monitoring of hydrostatic balance and anchoring depth.

These case studies highlight the essential role of mechanical cutters in modern well intervention. From safety-valve repairs to tubing recovery and abandonment, success depends on precise tool deployment, secure anchoring, torque management, and robust contingency planning. The next section details the equipment used, deployment and testing methods, and key diagnostics—such as drift runs—that confirm operational readiness.

Description and Application of Equipment and Processes

Overview of Planned Intervention Program

The Well A intervention was executed to retrieve a previously installed straddle that had failed inflow testing and to restore well integrity through a combination of mechanical cutting, staged fishing, and diagnostic logging. The operation was structured into discrete, risk-mitigated phases, each underpinned by rigorous verification and contingency protocols.

Pre-Operation Preparation and Barrier Assurance. The operation began with the mobilization of wireline and coiled tubing (CT) equipment, including Pressure Control Equipment (PCE), risers, and safety systems. All equipment underwent functional and pressure testing to 3,250 psi. Barrier integrity was confirmed through inflow and leak testing of the Christmas Tree (XMT) valves. A full hydrogen sulfide (H₂S) contingency plan was activated due to the presence of 571 ppm H₂S.

Comprehensive Toolbox Talks (TBTs) were conducted before each phase, and all temporary equipment was grounded to mitigate static discharge. The well was isolated from the process system, and wellhead valves were placed under manual override.

Milling of the tubing hanger. Initial coiled-tubing runs indicated that the straddle could be lifted to the tubing-hanger depth but could not pass through a 0.10-in. internal diameter (ID) restriction between the hanger and production tubing. Based on limited surface feedback, it was assumed that the straddle was positioned at the hanger with its fish-neck accessible for recovery.

To eliminate this restriction, a wireline-based milling solution was developed to enlarge the hanger bore to match the tubing ID. Following successful enlargement, the plan was to recover the straddle using a high-capacity electric-line stoker (rated to 100,000 lbf) combined with a grapple tool capable of positive latching and real-time surface read-out (SRO).

Access Verification and Inner Mandrel Drift Confirmation. Access to the straddle was confirmed through a sequence of drift runs using 3.855-in. gauge cutters and prong tools. Downhole video inspection verified the top of the straddle and identified any residual debris or obstructions. Mechanical drift runs with no-go prongs simulated the cutter's reach and confirmed that the tool string could extend 37.5 in. into the inner mandrel.

Precision Cutting of the Inner Mandrel. A custom-engineered slim mechanical cutter was deployed to sever the inner mandrel at the verified depth. Anchored by a hydraulic stoker, the cutter was actuated under continuous monitoring of current, torque, and piston travel via SRO. Circulation of inhibited seawater was maintained to manage swarf and pressure fluctuations. Indicators of a successful cut included:

- Drop in wireline tension
- Cutter piston travel confirmation
- Surface current profile analysis
- Post-run tell-tale measurements

Multi-Stage Fishing Operations. Following the cut, the upper section of the straddle was retrieved using the anchored stoker and grapple configuration. Controlled overpulls of up to 100,000 lbf were applied, with real-time monitoring for movement. Contingency plans were in place for re-cutting or switching to thermal cutting if required.

A magnet run was performed to remove residual slips and dogs before retrieving the lower section using the same stoker-grapple setup. Anchoring depths and pulling forces were carefully managed, and all movements were validated through tension monitoring and downhole pressure readings.

Diagnostic Logging and Integrity Verification. Post-retrieval, a series of diagnostic logs were performed:

- Downhole video survey to confirm the position of the tubing-retrievable surface safety valve (TRSSSV) flapper
- 40-arm Multifinger Caliper (MFC) log to assess tubing condition, revealing up to 70% wall penetration at 164.3 ft measured depth below rotary table (MDRT)
- Leak detection survey to identify tubing-to-annulus communication.

These diagnostics informed the final repair strategy and validated the success of the retrieval operation.

Repair and Restoration. The well was restored by installing a modular isolation straddle system and a Wireline-Retrievable Subsurface Safety Valve (WRSSSV) with high-integrity sealing elements. The lower packer was deployed on electric line, while spacer pipes and the upper packer were installed on slickline. All installations were pressure-tested and inflow-verified. A contingency WRSSSV with swellable seals was staged in case of sealing failure.

Contingency Planning and Risk Mitigation. The programme incorporated extensive contingencies, including:

- Thermal cutting if mechanical cutting failed
- CT milling assemblies for scale or mandrel removal
- Multiple fishing tool configurations for wireline and CT
- Jet-blasting runs for debris removal
- Real-time camera diagnostics for decision-making

Each phase was supported by detailed risk assessments and executed under strict adherence to well-service procedures and Health, Safety, and Environment (HSE) protocols.

The following section builds on the equipment descriptions and operational methodology outlined earlier, presenting the actual performance data, observations, and outcomes from each phase of the intervention.

Presentation of Data and Results

This section summarizes the key findings from each phase of the intervention, supported by figures, logs, and tables for clarity. Results are presented in a structured format to highlight operational performance, confirm tool functionality, and compare planned objectives with actual outcomes. Where applicable, real-

time surface read-out (SRO) data, diagnostic logs, and post-job inspections are referenced to validate the effectiveness of the milling, cutting, and staged fishing operations.

Milling of the Tubing Hanger

The tubing-hanger milling phase was successfully executed but presented several operational challenges and lessons learned:

- In-tree milling risk: Performing milling inside the Christmas tree introduced significant operational risk. The inclusion of contingency disconnect subs was critical to ensure tool recoverability in the event of sticking.
- Complex straddle design: The construction of the straddle and the potential for mechanical separation of the fish-neck complicated recovery attempts.
- Outcome vs. expectation: Despite successful enlargement of the tubing-hanger bore and confirmation of tool functionality during the Surface Integration Test (SIT), downhole conditions prevented complete straddle recovery.
- Diagnostic limitations: Limited feedback during earlier coiled-tubing operations led to assumptions regarding straddle position and restriction severity, which were later corrected through electric-line diagnostics.

Key Observations.

- The milling operation successfully restored the tubing-hanger ID to match the production tubing, enabling subsequent intervention steps. The design of the mill is shown in Fig. 2, and the profile which was milled in illustrated in Fig. 3.
- Real-time monitoring during milling (Fig. 4) confirmed stable torque and cutting performance, validating SIT findings.
- The inability to retrieve the straddle after hanger enlargement highlighted that the primary barrier was not the hanger restriction alone but the mechanical engagement of slips deeper in the completion.



Figure 2—Milling Assembly: NPR mill bit with integrated disconnect sub, designed for safe recovery in case of tool sticking.

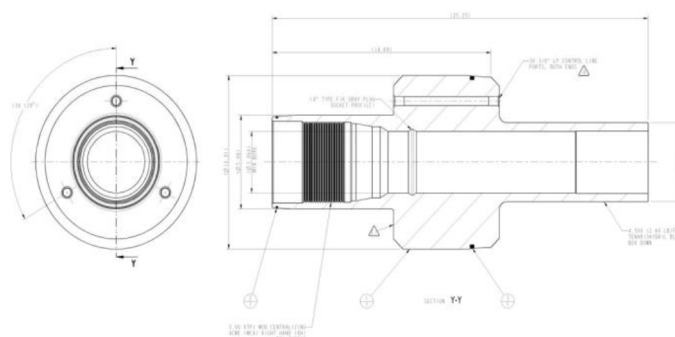


Figure 3—Tubing Hanger Profile: Pre- and post-milling geometry illustrating the removal of the 0.10-in. ID restriction.

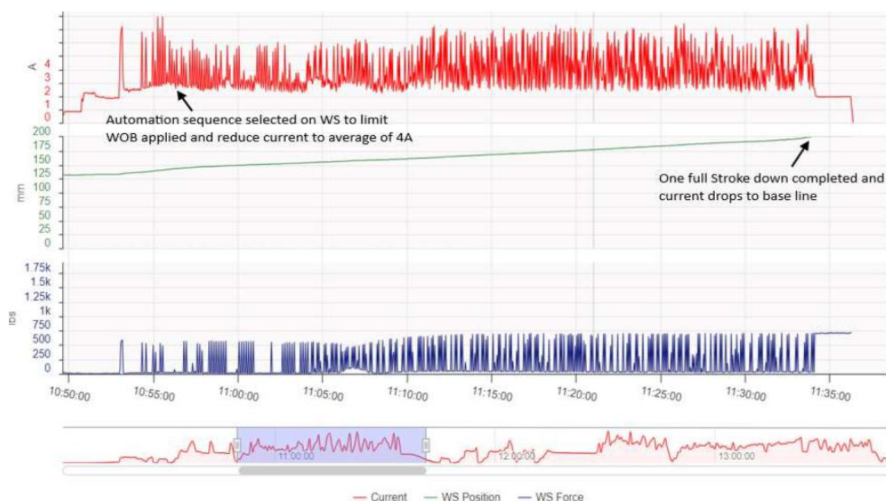


Figure 4—Log Data: Real-time milling log showing torque response and depth correlation during hanger enlargement.

Drift Checks

Drift check runs were a critical component of the Well A straddle retrieval program, serving as both a diagnostic and assurance mechanism to validate tool access, confirm internal diameters, and ensure operational readiness prior to high-risk mechanical interventions. These runs were strategically staged throughout the program to de-risk each subsequent phase and verify that the wellbore geometry was compatible with the planned bottomhole assemblies (BHAs).

Objectives of Drift Runs.

- **Confirm Access to the Top of Fish (TOF):** A 3.855-in. gauge cutter was deployed to confirm that the tubing was clear of debris and that tools could reach the top of the straddle at 1,866.4 ft measured depth below rotary table (MDRT). This ensured that the tubing hanger, previously milled to 3.958 in., provided sufficient clearance for retrieval tools.
- **Verify Inner Mandrel Accessibility:** A series of prong tools were used to simulate the outer diameter (OD) and extension length of the mechanical cutter. These tools confirmed that the cutter could be accurately positioned and extended to the target cutting depth of 37.5 in. within the inner mandrel.
- **Establish Mechanical Baselines:** Each drift run recorded running and pulling weights, identified abnormal resistance, and detected potential obstructions or scale accumulation. This data was essential for calibrating toolstring behavior and anchoring strategies during cutting and fishing phases.
- **Validate Toolstring Compatibility:** Drift runs confirmed that the planned cutting and fishing tools could be deployed without interference. Prong tools were dimensionally matched to the cutter's

OD (1.55 in.) and extension length, ensuring that the cutter arms would reach the intended cut window without risk of premature activation or misalignment.

- **Support Contingency Planning:** In the event of restricted access or unexpected obstructions, drift results provided the basis for activating contingency plans, such as switching to smaller OD tools, deploying magnets for debris retrieval, or initiating coiled-tubing milling operations.

Execution and Verification. Each drift run followed a standardized procedure:

- **Pre-job alignment:** Objectives and risk mitigation strategies were reviewed in pre-job meetings.
- **Fluid conditioning:** The well was bullheaded with inhibited seawater to maintain clean fluid conditions.
- **Controlled deployment:** The drift tool was run in hole (RIH) at a controlled speed, with real-time monitoring of weight and depth parameters.
- **Depth confirmation:** Upon reaching the target depth, the tool was carefully tagged against the no-go profile or inner mandrel.
- **Measurement validation:** The extension of the prong relative to the no-go was measured upon retrieval to verify that the cutter would achieve the intended depth. Special attention was given to the prong simulating cutter deployment, which featured a deliberate extension of 221.9 mm to replicate the cutter's reach.

This procedure confirmed that the cutter could be effectively anchored above the straddle and extended accurately into the mandrel without interference, reducing operational risk and ensuring readiness for the cutting phase.

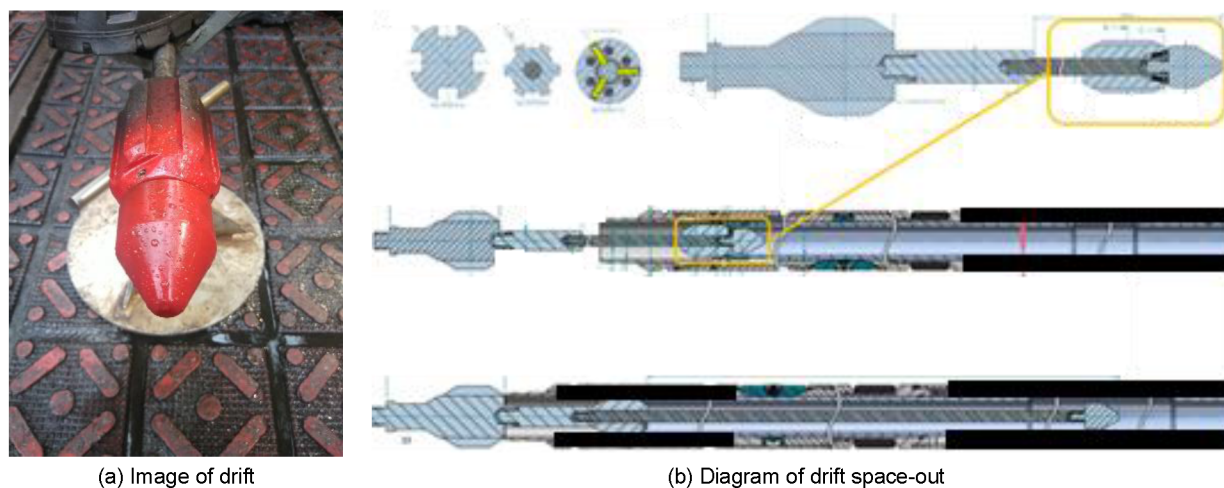


Figure 5—Illustrations of space out drift.

Surface Integration Testing of the Slim Mechanical Cutter for Inner Mandrel Segmentation

Following unsuccessful attempts to retrieve a stuck straddle in Well A—despite applying up to 100,000 lbf of axial force—a precision cutting solution was required to segment the inner mandrel and release slip loading. The proposed approach involved developing a slim-profile mechanical cutter capable of entering a 1.811-in. internal diameter (ID) and performing a controlled cut 975 mm inside a potentially unstable assembly. The objective of the Surface Integration Test (SIT) was to validate this solution under simulated downhole conditions before offshore deployment.

A rapid development program was initiated to design and qualify a cutting head that could:

- Pass through the restricted ID and accurately land at the target depth.
- Deliver a low-torque cutting action to minimize mandrel rotation.
- Generate minimal swarf for easier debris management.
- Provide positive depth confirmation via a mechanical tell-tale.

The cutting head was integrated with an electric-line powered rotary platform and configured for real-time monitoring of torque, current, and piston travel.

The system comprised:

- Slim mechanical cutting head (1.7 in.): Designed for restricted-ID applications, featuring grinding pads optimized for low-torque cutting and minimal swarf generation.
- Anchoring and stroking module: Provided axial stability and controlled feed during cutting.
- Surface control and monitoring unit: Enabled continuous observation of cutting parameters and depth verification.
- Test mandrel: A full-scale replica of the in-well geometry mounted horizontally beneath a 3½-in. tubing section, with continuous water circulation to simulate downhole fluid conditions.

The SIT was conducted in three runs to validate tool performance and optimize cutting parameters

- Run 1: Initial configuration with first-generation grinding pads. After two pressure adjustments, optimal cutting pressure was determined at 58 bar. Cut completed in under 2 hours. Findings prompted redesign of pad geometry for improved efficiency.
- Run 2: Second-generation pads with increased segment count and modified back angle. Cutting pressure raised to 62 bar; cut completed in approximately 2 hours under conservative torque conditions.
- Run 3: Same configuration as Run 2, but with reduced cutting pressure to simulate conservative field conditions. Cut completed in approximately 4 hours.

Between runs, the tool was fully redressed and inspected to confirm mechanical integrity.

All three cuts were successfully executed, confirming the tool's capability to perform precise, low-torque cuts in a restricted-ID mandrel. The grinding-based cutting method proved effective in minimizing swarf and reducing mechanical stress on the mandrel.

Key recommendations for field deployment.

- Maintain cutting pressure at approximately 62 bar, with pre-adjusted relief valves for fine-tuning.
- Continuously monitor Surface Read-Out (SRO) data (torque, current, piston travel) to track progress and detect anomalies.
- Prepare contingency for additional cuts: If required, a second cut can be initiated in the existing groove or at a new location based on SRO feedback.

The SIT validated the mechanical integrity, cutting efficiency, and operational readiness of the slim mechanical cutter for the Well A intervention (Fig. 6). This qualification step was critical in de-risking offshore execution under tight operational timelines.



Figure 6—Illustration of site testing of cutter prototype.

Results of the Inner Mandrel Cutting and Retrieval Offshore on Well A

Offshore deployment commenced with the slim mechanical cutter configured for restricted-ID operations. A mechanical telltale confirmed that the cutting arms reached the precise target depth, while real-time surface read-out (SRO) provided continuous feedback on torque, current, and piston travel throughout the operation. The mandrel cut was successfully completed within a few hours, with clear indicators of cut completion observed in the SRO data.



Figure 7—Cutting inner mandrel slim cutter.

Following the cut, the staged fishing program was executed using a high-capacity hydraulic stoker combined with an electrically actuated grapple tool. On the first run, a positive latch was confirmed, and the straddle assembly parted as planned, enabling recovery of the upper section. A downhole video camera was then deployed to visually inspect the new top of fish and verify wellbore conditions before proceeding.

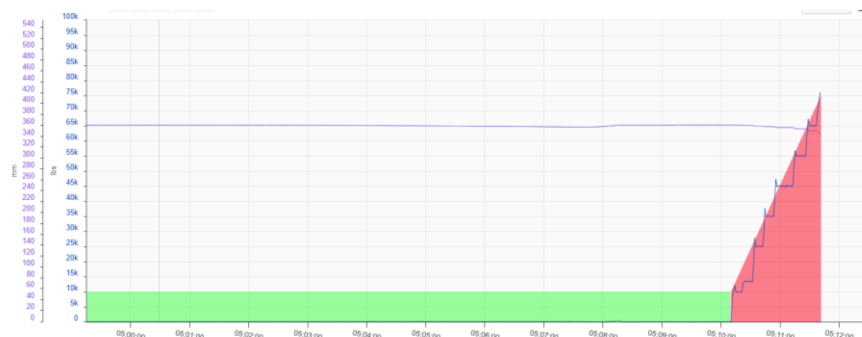


Figure 8—Retrieving straddle 75Klbs force to break.

The second run engaged and retrieved the middle section of the straddle using the same stroker-grapple configuration. A third run was subsequently performed to latch and recover the remaining lower section, completing the multi-stage recovery sequence. (Fig. 10).

Key Outcomes.

- Cutting performance: The internal mandrel cut was completed efficiently, with no tool stalls or unexpected torque spikes.
- Fishing success: All straddle segments were recovered in three controlled runs, minimizing risk of secondary intervention.
- Diagnostic assurance: Real-time SRO and downhole video provided critical confirmation of tool performance and wellbore conditions, enabling informed decision-making at each stage.

This operation demonstrated that a combination of thorough planning, purpose-built tooling, and real-time feedback systems can enable safe and efficient recovery of complex stuck assemblies. The ability to obtain reliable downhole data was essential for adapting the strategy during a technically challenging and time-sensitive intervention.

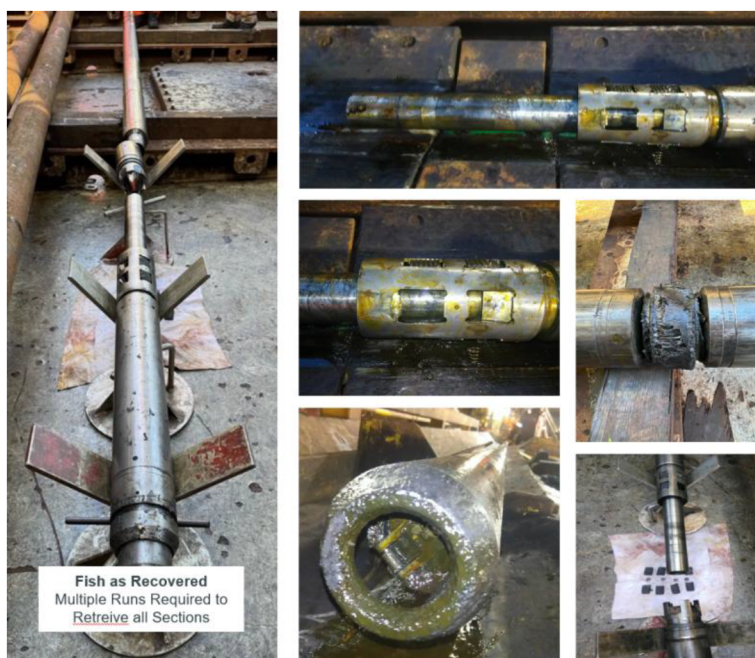


Figure 9—Images of recovered straddle.

Conclusions

The intervention in Well A successfully demonstrated the viability of precision electric-line cutting and multi-stage fishing operations within a severely restricted internal diameter (ID) environment. The custom-engineered slim mechanical cutter, designed for a 1.811-in. ID mandrel, performed reliably under both surface integration testing and offshore deployment, achieving controlled segmentation with minimal swarf and torque.

Real-time surface read-out (SRO) telemetry enabled dynamic decision-making, confirming tool engagement, cut completion, and fish movement. The staged recovery of the straddle assembly—executed with hydraulic stroker and grapple tools—validated the effectiveness of force ramping and anchored fishing in overcoming slip embedment and scale-induced confinement.

This case highlights the importance of adaptive intervention strategies that integrate diagnostics, bespoke tooling, and contingency planning. The operation restored full well access without explosives, minimized operational risk, and reinforced the value of agile engineering in time-critical well integrity scenarios. These findings offer practical guidance for similar interventions in mature offshore assets facing scale, corrosion, and mechanical constraints.

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Nomenclature

Symbol	Definition
BHA	Bottomhole Assembly
CRA	Corrosion-Resistant Alloy
CT	Coiled Tubing
DHSV	Downhole Safety Valve
EL	Electric Line
HPHT	High-Pressure High-Temperature
ID	Internal Diameter
klbf	Thousand Pounds Force
MDRT	Measured Depth Below Rotary Table
MFC	Multifinger Caliper
OD	Outer Diameter
PCE	Pressure-Control Equipment
RIH	Run In Hole
SCSSV	Surface-Controlled Subsurface Safety Valve
SIT	Surface Integration Test
SL	Slickline
SRO	Surface Read-Out
TOF	Top of Fish
TRSSSV	Tubing-Retrieveable Surface Safety Valve
WRSSSV	Wireline-Retrieveable Subsurface Safety Valve
XMT	Christmas Tree

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